Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.1 – Circuit Diagrams & Symbols

Unit 2: Electricity & Energy

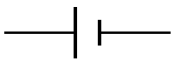
The question: *how do you tell someone else exactly how to build your circuit, without drawing a picture of every wire and battery?* Electricians, engineers and physicists all use the same trick: a small set of **symbols**, agreed everywhere, standing in for the real components. Learn the symbols and you can read — and build — any circuit in this unit.

Learning intentions

All students will learn the symbol, name and job of: wire, cell, switch, bulb, resistor — and build and draw a simple circuit from a diagram.
Most students will also use a **variable resistor** to control a circuit.
Some students will also meet the **LED** and the **capacitor**.

Part 1 — The symbols everyone needs

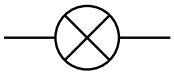
Five components, five symbols. Learn the shape, not just the name — in a diagram you will only ever see the shape.



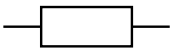
Cell
pushes charge round the circuit



Switch
opens and closes the loop



Bulb
lights up when current flows



Resistor
resists the flow of current

A plain straight line is the **wire** — there is no separate symbol for it, because the wire *is* the line joining everything else together.

1) **Label** the four symbols below with their names, from memory — no peeking at the row above.



Build it

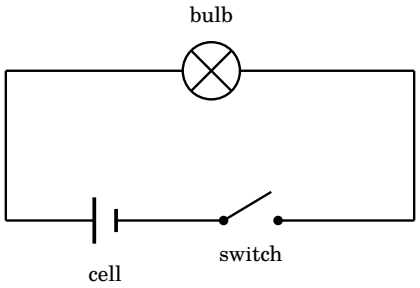
Take a cell, a switch, a bulb and three wires. Build the circuit so the bulb lights when the switch is closed. **Predict first, then test:** what happens to the bulb when you open the switch?

Check: the bulb lights with the switch closed, and goes out the instant you open it. If it does not, you have a gap in the loop — follow the wire all the way round and find it.

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Part 2 — Reading and drawing a circuit diagram

A **circuit diagram**, also called a **schematic diagram**, is the real circuit, redrawn as a neat loop of symbols. Here is the circuit you just built, drawn properly:



Schematic diagram rules

Write in each rule as we go through it together.

1 _____

_____

2 _____

_____

3 _____

_____

_____

Good circuit diagrams

- Use a ruler and a sharp pencil.
- Draw the loop as a neat rectangle.
- Keep every symbol roughly the same size.
- Follow the loop all the way round before you lift your pencil.

2) **Draw** the circuit diagram for: a cell, a closed switch, and **two** bulbs, one after another in a single loop. Use the rules above.

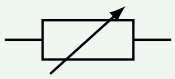
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Part 3 — Adding a resistor

A **resistor** does one job: it resists the flow of current, on purpose, so less current flows than without it.

3) **Predict** what happens to the brightness of a bulb if you add a resistor into the same loop, in series with it. Then explain *why*, using the word **resist**.

Most students — the variable resistor

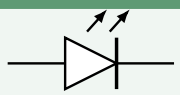


Variable resistor

a resistor you can turn up or down

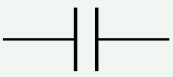
4) **Draw** the circuit diagram for: a cell, a closed switch, a bulb, and a variable resistor, all in one loop. Then state what turning the variable resistor's knob *up* does to the bulb.

Some students — LED and capacitor



LED

lights up, but current can only flow one way through it



Capacitor

stores charge, then releases it

5) **Explain** the one thing an LED can do in a circuit that an ordinary bulb cannot.

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Part 4 — On your own

A torch contains a cell, a switch, and a bulb, connected in a single loop.

6) **Draw** the torch's circuit diagram, labelling each symbol. Then add a **variable resistor** into the same loop, so the torch's brightness can be turned up and down.

HOMEWORK

1. Draw and label the symbol for each of the five components from Part 1, from memory.
2. Find **one** real circuit at home or at school (a torch, a lamp, a remote control …) that uses at least three of today's components. Name the object and list the components it uses.

Hand in: your five symbols and your one real example, next lesson.